AI Analog Circuit Design Agents : On Knowledge Extraction and Transfer with Knowledge Graphs

Karthik Somayaji N.S
Dept. of Electrical and Computer Engineering
University of California, Santa Barbara
karthi@ucsb.edu

Peng Li
Dept. of Electrical and Computer Engineering
University of California, Santa Barbara
lip@ucsb.edu

2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD) | 979-8-3315-1560-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCAD66269.2025.11240770

***Abstract*—Emulating expert human designers' cognitive strategies is a long-standing aspiration in analog design automation. A persistent challenge is achieving generalizability, specifically, how insights from optimizing certain circuits can effectively inform the design of related circuits with differing topologies. For the first time, we identify required key components towards enabling AI-based topology-agnostic design knowledge transfer and demonstrate how such methodology can be developed for analog sizing optimization utilizing large language models (LLMs). The proposed framework, called Analog Design Optimization with Knowledge Transfer (`ADO-KT`) consists of three key integrated steps: 1) LLM-enabled knowledge graph (KG) generation for structured extraction of prior analog design knowledge, 2) LLM-enabled knowledge graph refinement for quality KG generation, and 3) topology-agnostic knowledge transfer for LLM-assisted optimization of new circuits. `ADO-KT` enables seamless summarization and transfer of quality knowledge flow between circuits of varying types and topologies, effectively mirroring human designers' strategies. Empirical evaluations demonstrate significant improvements in design quality and data efficiency, highlighting the potential of the proposed approach for automated analog circuit design with generalized knowledge reuse.**

***Index Terms*—analog design, LLM assisted design, knowledge graphs, knowledge graph filtering**

I. Introduction

Analog circuit design remains central to modern integrated circuits (ICs), directly impacting performance, power efficiency, and reliability. Unlike digital design, analog design demands careful navigation of continuous, interdependent relationships between circuit topologies, design parameters, and performance metrics [1], [2]. Traditionally, this process has relied heavily on human expertise, with designers drawing upon accumulated experience and intuition to optimize complex analog blocks.

Recent progress in analog design automation has introduced algorithmic approaches such as genetic algorithms [1], [3], Bayesian Optimization (BO) [2], [4], reinforcement learning (RL) [5], [6], [7], [8] and LLM based solutions [9], [10], [11], [12] to explore design spaces more systematically. In particular, GCN-RL [6] enables knowledge transfer across circuits by leveraging graph convolutional networks within an RL framework. However, GCN-RL relies on black-box RL optimization rather than interpretable, cognitive reasoning via language models. Its transfer is limited to circuits with similar functionalities and assumes architectural compatibility between the circuit graph and the GCN structure, making it difficult to generalize across circuits with significantly differing or more complex topologies.

This motivates a central research question: *How can design knowledge be effectively transferred across analog circuits that share only partial structural or functional similarities?*

Consider the task of optimizing a low dropout regulator (LDO), which incorporates amplifier sub-blocks. Existing analog EDA techniques typically require an exact architectural match with a previously optimized LDO to enable knowledge reuse. In contrast, human designers often draw on insights from prior sub-block-level designs—such as amplifiers—even when there is only partial topological or functional alignment. This flexible reuse of design experience reflects the adaptive reasoning of expert engineers and highlights a fundamental limitation of existing automated methods.

To emulate this flexibility, we present the first work on analog sizing optimization with topology-agnostic knowledge transfer. In the proposed framework called Analog Design Optimization with Knowledge Transfer (`ADO-KT`), we establish a structured knowledge representation that captures generalizable design insights beyond rigid topology matching. A desirable framework should encode the complex causal relationships among performance metrics, substructures, and design parameters—an area where knowledge graphs (KGs) have shown promise across domains involving causal reasoning [13], [14]. Motivated by this, we introduce a hierarchical KG-based formalism to represent analog design knowledge. Since manually constructing analog-specific KGs is impractical, we develop an automated methodology that uses hierarchical prompting with Large Language Models (LLMs) to generate these representations efficiently.

Yet, generating KGs is only part of the solution. To integrate structured knowledge into the optimization of new circuits, we embed hierarchical KGs into a hybrid LLM and Bayesian optimization (BO) loop. This allows this LLM-assisted optimization process to dynamically incorporate prior knowledge from structurally or functionally related circuits, guiding the search for optimal design points. Unlike recent LLM-based approaches such as [9], [10], which treat each circuit optimization task independently, our framework explicitly supports cross-circuit knowledge reuse via structured context injection.

To ensure robustness in knowledge extraction, we employ consistency-based filtering to discard unreliable or contradictory knowledge entries, enhancing trust in LLM-generated KGs. Our `ADO-KT` approach enables practical, generalizable knowledge transfer across circuits without requiring architectural equivalence—closely emulating expert design intuition.

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

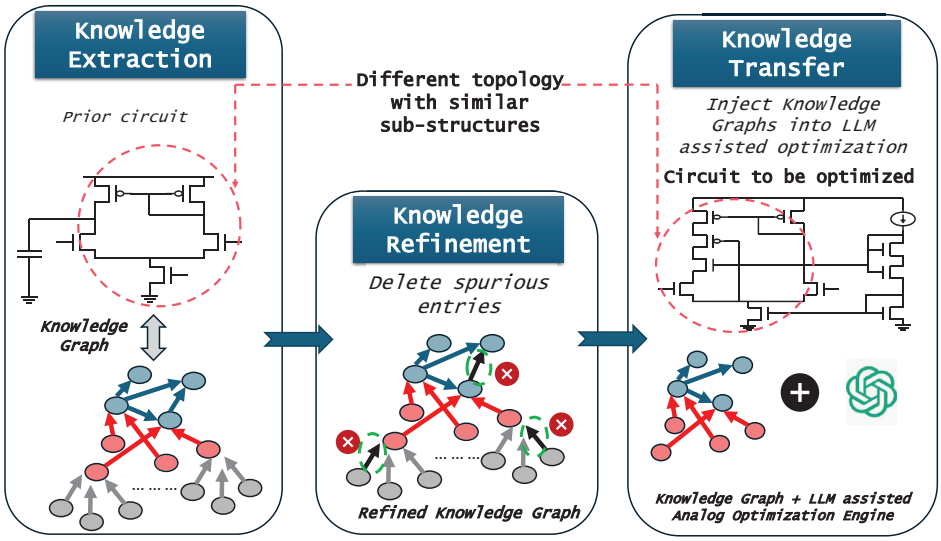


Fig. 1: How can we transfer knowledge between circuits with different topologies ? Our methodology consists of constructing **Knowledge Representations** for related circuits using knowledge graphs, refining such knowledge graphs using our **Knowledge Refinement** mechanism and finally using a LLM assisted optimization framework for **Knowledge Transfer**

Our key contributions include:

1) **Structured Knowledge Extraction:** A hierarchical LLM-based prompting approach to systematically generate analog-specific knowledge graphs (KG) encoding essential causal relationships.
2) **Robust Knowledge Graph Refinement:** Consistency-driven filtering techniques ensuring reliability and precision of generated knowledge graphs.
3) **Topology-Agnostic Knowledge Transfer:** A knowledge graph infused hybrid BO-LLM optimization framework that integrates structured knowledge across varying types of circuits, emulating human-like knowledge transfer flexibility.

The proposed ADO-KT framework is illustrated in Figure 1. Extensive experiments confirm substantial improvements in design quality and sample efficiency, highlighting the effectiveness of structured knowledge representation and transfer in automated analog circuit optimization.

II. Transferrable Structural Analog Knowledge Representation

A. Structural Analog Design Knowledge

Given a circuit topology, analog circuit design involves selecting and tuning design parameters (sizing values) while trading off among multiple competing performance metrics (design objectives). Design knowledge, in this context, refers to a structured understanding of the causal interdependencies among design parameters, subcircuit structures, and the performance metrics. This knowledge can be systematically organized along three key dimensions:

(1) **Interactions between Performance Metrics:** A key challenge in analog design stems from its multi-objective nature. For this, understanding the intrinsic relationships between various performances such as trade-offs or direct proportionality is essential. Such relationships offer reasoning about the impact of improving one metric on others.

(2) **Dependence on Topology:** Specific sub-structures within the circuit topology tend to exert dominant influence over certain performance metrics. Recognizing these associations enables a principled understanding of how topology of sub-circuits shapes circuit behavior through certain performance metrics. Since our goal is to enable topology-agnostic knowledge transfer across circuits, it is essential to capture how individual sub-circuit blocks influence performance. This is particularly motivated by the observation that many analog circuits share recurring substructures, allowing such localized knowledge to generalize across different larger circuit topologies.

(3) **Structure in Design Parameters:** Design parameters can be categorized based on their association with sub-structures. The influence of each parameter on performance is typically mediated through the sub-structure to which it belongs to.

These structural perspectives form the basis for systematically representing and reasoning about the causal nature of analog design knowledge.

B. Exacting Design Knowledge Using Knowledge Graphs

Inspired by the inherently causal nature of analog design processes, we propose using causal graphs, specifically, Knowledge Graphs (KGs) [15]–[17] to explicitly capture the complex interdependencies among performance metrics, circuit substructures, or design parameters. Formally, a KG for a given circuit C_k is defined as:

$$G_k = (V_k, E_k), \tag{1}$$

where V_k denotes nodes representing key entities such as performance metrics $P_i \in \mathcal{P}$, circuit substructures $S_i \in \mathcal{S}$, and design parameters $x_i \in \mathcal{X}$. Directed edges $E_k \subseteq V_k \times V_k$ characterize relationships between these entities. Each edge

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

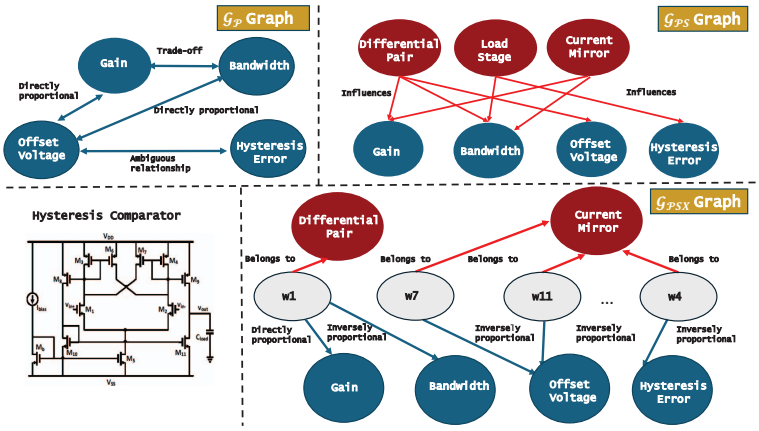


Fig. 2: Pictorial representation of some sample entries of the KG for the Hysteresis Comparator generated by LLM ($\mathcal{M}_{\text{KG}}$: LLama3-70B-instruct) using the proposed hierarchical approach.

$e = (v_i, r, v_j) \in E_k$ connects nodes v_i and v_j, with r describing the relationship type. This structured representation offers an efficient way for interpreting and representing structural and functional knowledge integral to the analog design process.

C. Hierarchical Analog Design Knowledge Graphs: Overview

The absence of readily available analog-specific knowledge graphs, coupled with the substantial manual effort required for their construction, presents a key bottleneck in knowledge transfer driven design. To address this, we introduce a hierarchical methodology that leverages Large Language Models (LLMs) to automatically generate analog design knowledge graphs (KGs). An illustration of our framework to construct KGs is shown in Figure 3.

Our approach constructs the KG in a staged, hierarchical fashion—progressively capturing the relationships among performance metrics, circuit substructures, and design parameters. A representative example of some sample entries of the resulting KG generated by our approach using a sample LLM (LLama3-70B-instruct) is illustrated in Figure 2. The KG is constructed sequentially and consists of the following three subgraphs:

(1) **Performance Metric Graph ($\mathcal{G}_\mathcal{P}$):** Captures the interactions among performance metrics $\mathcal{P}$, such as trade-offs and proportional dependencies. For instance, in Figure 2, $\mathcal{G}_\mathcal{P}$ identifies a trade-off relationship between gain and bandwidth.

(2) **Performance–Structure Graph ($\mathcal{G}_{\mathcal{PS}}$):** Maps each performance metric to the substructures $\mathcal{S}$ that predominantly influence it. As shown in the example, the differential pair significantly influences both gain and bandwidth.

(3) **Performance–Structure–Parameter Graph ($\mathcal{G}_{\mathcal{PSX}}$):** Encodes how design parameters $\mathcal{X}$, associated with specific substructures, affect the corresponding performance metrics. For example, parameter w_1 within the differential pair exhibits a direct proportional relationship to the gain.

The complete knowledge graph G_k for a circuit C_k is formed by unifying the three subgraphs:

$$G_k = \mathcal{G}_\mathcal{P} \cup \mathcal{G}_{\mathcal{PS}} \cup \mathcal{G}_{\mathcal{PSX}}.$$

This hierarchical construction enables a structured and interpretable representation of analog design knowledge, serving as a foundation for interpretable knowledge transfer.

Our approach constructs the KG for a given circuit C_k by querying a language model $\mathcal{M}_{\text{KG}}$ using a hierarchical querying procedure. Each sub-graph is generated sequentially, with each stage leveraging the results of the previous one, along with a shared circuit context. This process is expressed as:

$$\begin{aligned}
\mathcal{G}_\mathcal{P} &\leftarrow \mathcal{M}_{\text{KG}}(\text{circuit context},\ \mathcal{G}_\mathcal{P}\ \text{query}) \\
\mathcal{G}_{\mathcal{PS}} &\leftarrow \mathcal{M}_{\text{KG}}(\text{circuit context},\ \mathcal{G}_\mathcal{P},\ \mathcal{G}_{\mathcal{PS}}\ \text{query}) \\
\mathcal{G}_{\mathcal{PSX}} &\leftarrow \mathcal{M}_{\text{KG}}(\text{circuit context},\ \mathcal{G}_\mathcal{P},\ \mathcal{G}_{\mathcal{PS}},\ \mathcal{G}_{\mathcal{PSX}}\ \text{query})
\end{aligned}$$

We next describe the circuit context and the hierarchical querying procedure used for generating the sub-graphs.

D. Circuit Context

To enable context-aware generation, we first provide the LLM $\mathcal{M}_{\text{KG}}$ with the netlist of circuit C_k, which conveys the circuit's topological structure. Additionally, we supply representative design instances obtained from prior optimization runs, which reveal empirical relationships between performance metrics and design parameters. We also provide the specifications that were provided for the previous optimization runs of the circuit (for example - Gain $>$ 40dB, hysteresis error $<$ 0.2V etc). Together, this information along with the hierarchical queries allows the model to infer the underlying causal interactions that govern circuit behavior. This enables the generation of interpretable design knowledge.

E. Constructing the $\mathcal{G}_\mathcal{P}$ Graph

The first step in our hierarchical KG generation process in `ADO-KT` is the construction of the performance metric graph $\mathcal{G}_\mathcal{P}$. In this graph, nodes correspond to the performance metrics $\mathcal{P}$ relevant to the given circuit C_k. The goal is to capture the qualitative relationships among the metrics. The relationships permitted between performance metric nodes include:

- `trade-off` — an improvement in one metric is typically accompanied by degradation in another,
- `directly proportional` — both metrics tend to improve or degrade together,

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

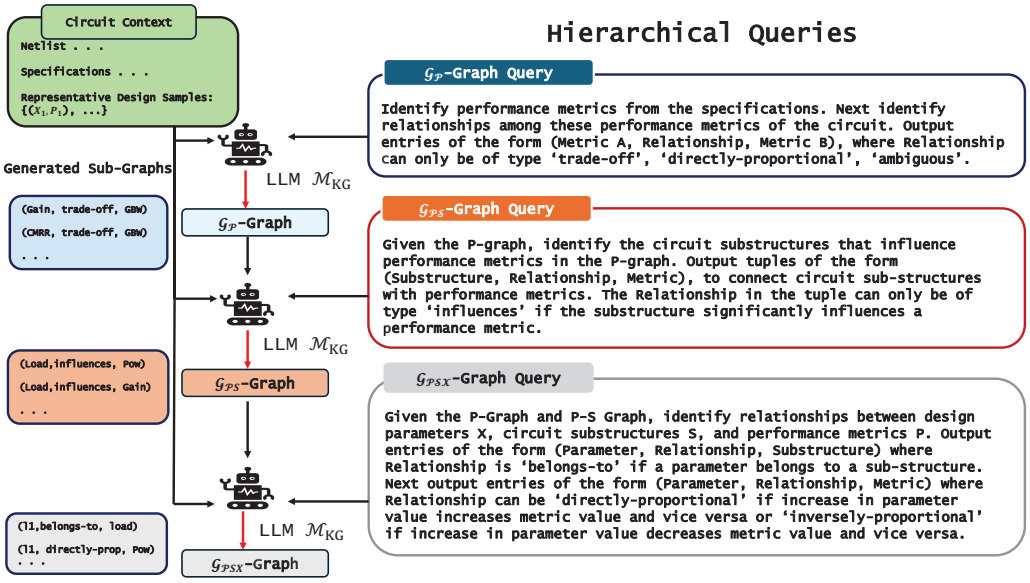


Fig. 3: Hierarchical Generation of the Knowledge Graph

- `ambiguous relationship` — the relationship between the metrics is unclear or context-dependent.

To construct $\mathcal{G}_{\mathcal{P}}$, we prompt the LLM $\mathcal{M}_{\text{KG}}$ with the $\mathcal{G}_{\mathcal{P}}$-**Graph query** (shown in Figure 3) to first identify the relevant performance metrics from the circuit context and design specifications - and then determine pairwise relationships between them using only the predefined categories above.

F. Constructing the $\mathcal{G}_{\mathcal{PS}}$ Graph

Given the constructed $\mathcal{G}_{\mathcal{P}}$, we proceed to build the performance–structure graph $\mathcal{G}_{\mathcal{PS}}$. In addition to the performance metrics from $\mathcal{G}_{\mathcal{P}}$, this graph includes new nodes corresponding to the circuit substructures $\mathcal{S}$, which are automatically extracted from the circuit netlist.

The possible relationship in this graph is defined as:

- `influences` — indicating that a particular substructure has a significant impact on a specific performance metric.

The LLM $\mathcal{M}_{\text{KG}}$ is queried using the $\mathcal{G}_{\mathcal{PS}}$ **Graph query** (shown in Figure 3) to identify additional nodes that are a substructure of the circuit topology and link them to relevant performance metrics using the `influences` relationship.

G. Constructing the $\mathcal{G}_{\mathcal{PSX}}$ Graph

The final stage involves constructing the performance–structure–parameter graph $\mathcal{G}_{\mathcal{PSX}}$, using $\mathcal{G}_{\mathcal{P}}$ and $\mathcal{G}_{\mathcal{PS}}$ sub-graphs as inputs. This graph augments the existing nodes with design parameters $\mathcal{X}$, such as transistor widths and lengths which are also extracted from the circuit netlist. Here two types of relationships between nodes are introduced:

- `belongs to` — connecting a design parameter to the substructure it is associated with,
- `directly proportional` or `inversely proportional` — expressing the nature of the design parameter's influence on a performance metric.

Here, the LLM $\mathcal{M}_{\text{KG}}$ is queried using the $\mathcal{G}_{\mathcal{PSX}}$ **Graph query** (shown in Figure 3) to identify the relevant design parameters and establish both structural relationships (parameter–substructure) and functional relationships (parameter–performance metric).

III. Knowledge Transfer for New Circuit Optimization with ADO-KT

A core motivation for emulating human design strategies is the ability to reuse knowledge from one circuit to guide the optimization of another, even when their topologies differ. While knowledge transfer can be facilitated by feeding our extracted analog design knowledge graphs into any suitable LLM-based design agent, we have found that the recently developed hybrid Bayesian Optimization and LLM framework ADO-LLM [9] is a good choice. ADO-LLM combines the principled exploration of BO with the zero/few shot-learning capabilities of LLMs. BO proposes new design points using a built-in Gaussian process (GP) model with quantified uncertainty while the LLM, referred to as a backbone-LLM $\mathcal{M}_{\text{backbone}}$ generates additional suggestions based on circuit specifications and few-shot examples. These components work in tandem to optimize a given new circuit using a fairly small amount of circuit-specific simulation data.

Although $\mathcal{M}_{\text{backbone}}$ offers broad general-purpose knowledge, it lacks the circuit-specific insights necessary for transferring optimization strategies across different circuit topologies—particularly at the level of sub-blocks or substructures. To bridge this gap, we enrich the LLM's context with structured knowledge graphs (KGs) derived from previously optimized, structurally related circuits. These KGs encode causal relationships among performance metrics, substructures, and design parameters, enabling effective knowledge transfer even in the absence of exact topology matches.

At each optimization iteration in `ADO-KT` the backbone-LLM $\mathcal{M}_{\text{backbone}}$ is prompted with a combined context com-

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

prising: (i) the circuit netlist and specifications, (ii) best performing design points, and (iii) **KGs $\{G_i\}$ from related circuits. This enriched context in ADO-KT enables $\mathcal{M}_{\text{backbone}}$ to generate design suggestions that are grounded in the current task while benefiting from transferable insights**. We illustrate the ADO-KT framework in Algorithm 1. This KG-infused BO-LLM framework mirrors the way human designers apply experience across designs, significantly enhancing both sample efficiency and design point quality.

Algorithm 1 Knowledge Transfer using ADO-KT

Input: Target circuit C_k, Knowledge graphs from related circuits $\{G_1, G_2, \ldots, G_{k-1}\}$, backbone-LLM: $\mathcal{M}_{\text{backbone}}$, KG-generative LLM model: $\mathcal{M}_{\text{KG}}$

Output: Optimized design parameters $\mathbf{x}_k^*$ and KG G_k

a. Initialize dataset $\mathcal{D}_k = \{(\mathbf{x}_j, f(\mathbf{x}_j))\}_{j=1}^n$ with n evaluated design points for circuit C_k

b. Construct combined context $\mathcal{C}_k$ for the backbone-LLM, consisting of:
- Circuit netlist and specifications for C_k
- Best few-shot examples from $\mathcal{D}_k$
- **Knowledge graphs $\{G_i\}_{i=1}^{k-1}$ from related circuits**

c. Repeat until convergence or budget is exhausted:
i. Use GP-based BO to propose design point $\mathbf{x}_{\text{BO}}$
ii. Get $\mathbf{x}_{\text{LLM}} \leftarrow \mathcal{M}_{\text{backbone}}(\mathcal{C}_k)$
iii. Simulate $f(\mathbf{x}_{\text{BO}})$ and $f(\mathbf{x}_{\text{LLM}})$, evaluate performance.
iv. Update dataset: $\mathcal{D}_k \leftarrow \mathcal{D}_k \cup \{(\mathbf{x}_{\text{BO}}, f(\mathbf{x}_{\text{BO}})), (\mathbf{x}_{\text{LLM}}, f(\mathbf{x}_{\text{LLM}}))\}$

d. Return best design point:

$$\mathbf{x}_k^* = \arg\max_{\mathbf{x} \in \mathcal{D}_k} f(\mathbf{x})$$

e. Return knowledge graph G_k obtained from $\mathcal{M}_{\text{KG}}$

IV. Analog Knowledge Refinement

While knowledge graphs generated by LLMs offer a powerful means of representing design knowledge, they may occasionally contain inconsistencies or ambiguous entries due to the underlying LLM's generation process. The adopted hybrid ADO-LLM approach mitigates potential hallucinations of the LLM [9] because of incorporation of the Bayesian optimization agent, which also reduces the negative influences of imprecise knowledge graphs. Nevertheless, the quality of knowledge directly still impacts the effectiveness of knowledge transfer. To address this, we propose a sub-graph-wise refinement technique to refine the knowledge graph before it is used in the ADO-KT framework for knowledge transfer. We detail the knowledge refinement mechanism in Algorithm 2 and illustrate it in Figure 4.

Our method aims to quantify the uncertainty of the LLM-enabled KG generation process. It begins by generating multiple knowledge graphs for a given circuit using repeated LLM $\mathcal{M}_{\text{KG}}$ invocations, each consisting of a $\mathcal{G}_{\mathcal{P}}$-Graph, $\mathcal{G}_{\mathcal{PS}}$-Graph, and $\mathcal{G}_{\mathcal{PSX}}$-Graph. One of these KGs is arbitrarily designated as the anchor KG, while the remaining KGs serve as reference KGs. We then evaluate the consistency of each entry in the anchor KG against matching entries in the reference

Algorithm 2 Knowledge Refinement

Input: Generated KGs $\{G_k^1, \ldots, G_k^M\}$ for circuit C_k, rejection threshold τ

Output: Refined KG G_k^{refined}

a. Select one KG G_a as the anchor KG; treat the others $\{\mathcal{G}_r\} = \{G_k^1, \ldots, G_k^M\} \setminus \{G_a\}$ as reference KGs.

b. For each sub-graph $g \in \mathcal{G}_{\mathcal{P}}, \mathcal{G}_{\mathcal{PS}}, \mathcal{G}_{\mathcal{PSX}}$:
i. For each entry $e \in g$:
a. Initialize `contradiction_count` $\leftarrow 0$, and `match_count` $m \leftarrow 0$.
b. For each reference KG $G_r \in \{\mathcal{G}_r\}$:
– For each entry $e_r \in G_r^g$:
• If e_r and e have matching entities:
– Increment $m \leftarrow m + 1$
– If e contradicts e_r:
∗ Increment `contradiction_count`
c. Compute rejection rate: $\gamma =$ `contradiction_count`$/m$ (if $m > 0$)
d. If $\gamma > \tau$, discard entry e

c. Return the refined KG G_k^{refined} containing surviving entries from G_a

KGs. If an anchor entry frequently contradicts corresponding entries in the references, i.e., it has a rejection rate γ greater than a rejection threshold τ, it is flagged as unreliable and is discarded.

This filtering step systematically removes inconsistent entries from the KG. The refined KG is subsequently used as part of ADO-KT to ensure that only high-confidence knowledge informs the design suggestions. We illustrate the average rejection rate of the knowledge entries generated by different LLMs ($\mathcal{M}_{\text{KG}}$) for different circuits in Figure 5. From our knowledge refinement mechanism, we find that some LLMs generate initial consistent KGs (evident by smaller rejection percentages), while other smaller LLMs (with less than 70 billion parameters) may generate inconsistent KG entries within the KG (evidenced by higher rejection percentages).

V. Experimental Evaluation

The objective of the developed topology-agnostic knowledge transfer method ADO-KT is twofold: (i) improve final design quality, and (ii) enhance data efficiency by accelerating convergence. We evaluate our framework on five analog circuits shown in Fig. 6), which demonstrates effective structural knowledge transfer across different types of circuits with varying topologies.

We benchmark against two baselines: (i) GP-based Bayesian Optimization (GP-BO) [2], and (ii) ADO-LLM [9], which applies BO-LLM optimization without knowledge transfer. Performance and sample efficiency are assessed using both the final Figure of Merit (FOM), computed as a weighted combination of normalized metrics, and its evolution as a function of data consumption respectively.

Each optimization begins with 5 initial zero-shot design points from the backbone LLM $\mathcal{M}_{\text{backbone}}$, followed by 20

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

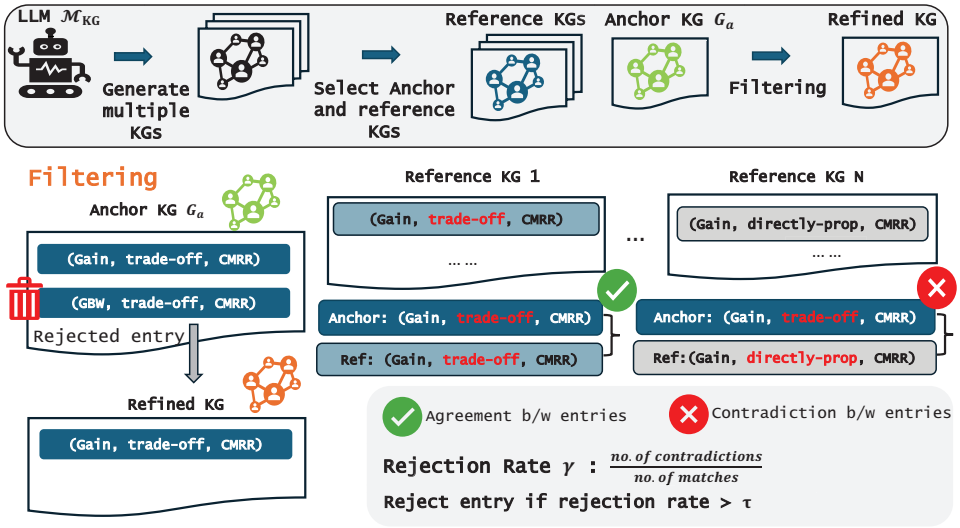


Fig. 4: Knowledge Refinement Framework: Every entry in anchor KG is checked for consistency with every matching entry of each reference KG.

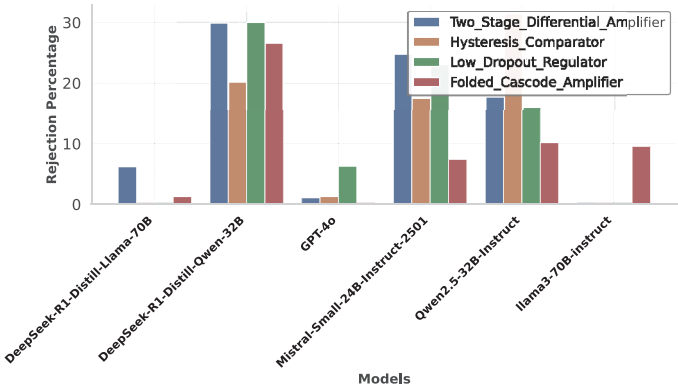


Fig. 5: Average rejection rate of the knowledge entries for different LLMs ($\mathcal{M}_{\text{KG}}$) for each circuit example

iterations where one candidate is proposed by $\mathcal{M}_{\text{backbone}}$ and one by BO, totaling 45 simulations per circuit (65 for the Hysteresis Comparator with 30 iterations).

All the experiments with ADO-LLM [9] and `ADO-KT` we use $\mathcal{M}_{\text{backbone}}$: GPT-3.5 [18] in the BO-LLM optimization loop. However, in order to generate KG, in `ADO-KT`, we employ a different LLM ($\mathcal{M}_{\text{KG}}$). To show the generality of the proposed techniques, we employ six different LLMs as candidates for ($\mathcal{M}_{\text{KG}}$) and tabulate optimization results in Tables I, II, III for each circuit. We use DeepSeek-R1-Distill-Llama-70B [19], DeepSeek-R1-Distill-Qwen-32B [19], GPT-4o [20], llama3-70B-Instruct [21], Mistral-Small-24B-Instruct-2501 [22] and Qwen2.5-32B-Instruct [23] as different choices for $\mathcal{M}_{\text{KG}}$ to generate the KG. We also compare performance with and without knowledge refinement to assess the role of KG consistency.

For knowledge refinement, we apply a conservative rejection threshold of $\tau = 0.3$. We begin with the two-stage differential amplifier (TSDA) to establish a baseline. As no prior circuits precede it, we optimize it using the ADO-LLM framework without knowledge transfer and generate its knowledge graph thereafter using LLM ($\mathcal{M}_{\text{KG}}$). Subsequent experiments evaluate topology-agnostic transfer across circuits.

A. Folded Cascode Amplifier (FC)

The folded cascode amplifier, which comprises 32 design parameters, contains functional substructures that are similar but not identical to those of the two-stage differential amplifier (TSDA). We inject the KG of the differential amplifier into the optimization process for the folded cascode circuit whose design specs are: gain $>$ 60 dB, CMRR $>$ 80 dB, and UGF $>$ 1 MHz. As shown in Table I, knowledge transfer using KGs with or without refinement improves overall performance while the proposed KG refinement leads to more metrics meeting specification constraints and achieves even greater FOM improvements compared to ADO-LLM.

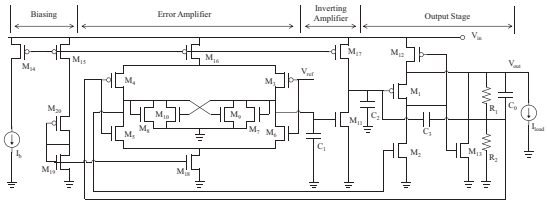
B. Low Dropout Regulator (LDO)

We evaluate our framework on a low dropout regulator (LDO) tasked with maintaining a stable 0.6 V output. The circuit spans 40 design parameters across 20 transistors and includes substructures such as error amplifiers and biasing blocks, similar to both the two-stage differential amplifier and the hysteresis comparator. The design constraints require a quiescent current below 12 mA and a regulated voltage error within 0.1 V of the target. As shown in Table II, KG-based transfer improves the FOM over ADO-LLM, with refined KGs yielding further gains.

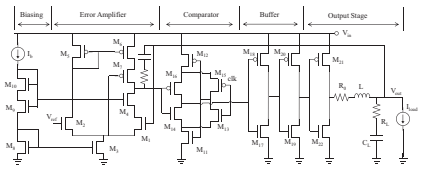
C. Hysteresis Comparator (Comp)

The hysteresis comparator, used in high-speed switching applications, consists of 24 design parameters with the following design specifications: gain $>$ 40 dB, bandwidth $>$ 0.1 MHz, offset error $<$ 20 mV, and hysteresis error $<$ 0.4 V. It contains key substructures—such as gain stages and biasing blocks which also appear in the two-stage differential amplifier. The structured KGs derived from the differential amplifier are

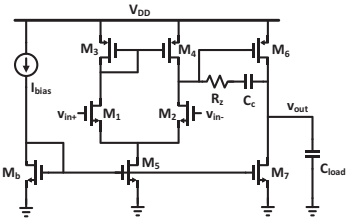
Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.



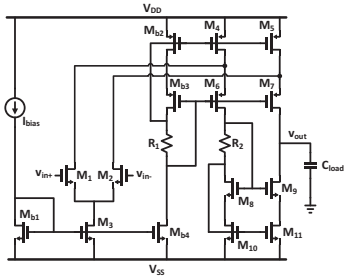
(a) Low Dropout Regulator



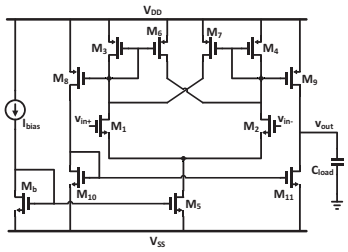
(b) DCDC Converter



(c) Two Stage Diff Amplifier



(d) Folded Cascode Amplifier



(e) Hysteresis Comparator

Fig. 6: Circuit Schematics.

TABLE I: Folded Cascode Amplifier Optimization Results

Method	Gain ↑ (Spec: > 60dB)	CMRR ↑ (Spec: >80dB)	GBW ↑ (Spec: >1MHz)	Power ↓ (Spec: <50 μW)	FOM ↑
GP-BO	4.96	55.54	1.439	65.86	-0.200
ADO-LLM	43.16	73.87	1.786	306.13	-0.102
ADO-KT : Knowledge Transfer (from TSDA) without Refinement with different $\mathcal{M}_{\text{KG}}$					
DeepSeek-Distill-Llama-70B	61.69	69.89	2.347	60.79	0.351
DeepSeek-R1-Distill-Qwen-32B	43.153	80.53	2.131	50.23	0.214
GPT-4o	65.58	89.79	2.054	54.12	0.364
Llama3-70B-instruct	66.19	88.41	2.154	54.12	0.380
Mistral-Small-24B-Instruct-2501	28.25	47.34	1.612	37.45	-0.018
Qwen2.5-32B-Instruct	45.24	96.42	1.726	259.6	-0.043
ADO-KT : Knowledge Transfer (from TSDA) with Refinement with different $\mathcal{M}_{\text{KG}}$					
DeepSeek-Distill-Llama-70B	65.18	87.84	2.215	55.27	**0.389**
DeepSeek-R1-Distill-Qwen-32B	68.26	91.55	1.934	52.06	0.352
GPT-4o	66.75	89.89	2.132	54.04	**0.386**
Llama3-70B-instruct	64.49	87.23	2.230	56.51	**0.388**
Mistral-Small-24B-Instruct-2501	61.80	84.25	2.284	59.98	0.382
Qwen2.5-32B-Instruct	60.617	91.38	1.921	54.09	0.321

TABLE II: LDO Optimization Results

Method	Qui Curr ↓ (Spec: < 12mA)	Reg. Voltage Error ↓ (Spec: < 0.1V)	FOM ↑
BO	5.018	0.466	0.064
ADO-LLM	5.011	0.457	0.055
ADO-KT : Knowledge Transfer (from TSDA, Comp) without Refinement with different $\mathcal{M}_{\text{KG}}$			
DeepSeek-Distill-Llama-70B	13.57	0.028	0.294
DeepSeek-R1-Distill-Qwen-32B	5.011	0.417	0.089
GPT-4o	30.66	0.068	0.166
Llama3-70B-instruct	10.05	0.0169	0.340
Mistral-Small-24B-Instruct-2501	5.011	0.457	0.055
Qwen2.5-32B-Instruct	5.011	0.457	0.055
ADO-KT : Knowledge Transfer (from TSDA, Comp) with Refinement with different $\mathcal{M}_{\text{KG}}$			
DeepSeek-Distill-Llama-70B	12.97	0.0285	**0.301**
DeepSeek-R1-Distill-Qwen-32B	14.8	0.0339	0.280
GPT-4o	12.0	0.021	**0.315**
Llama3-70B-instruct	8.31	0.0134	**0.365**
Mistral-Small-24B-Instruct-2501	5.03	0.3402	0.152
Qwen2.5-32B-Instruct	23.38	0.083	0.185

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

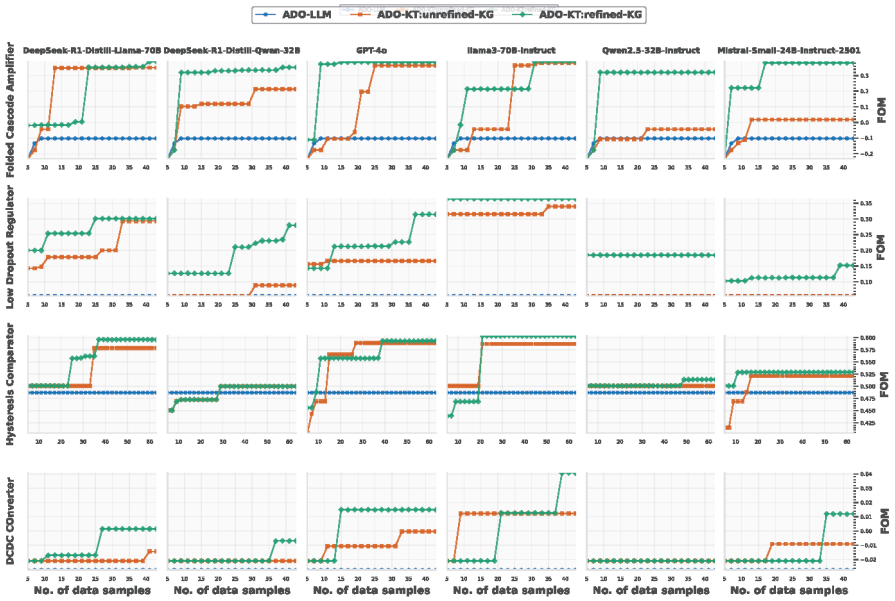


Fig. 7: Assessing Sample Efficiency: The evolution of FOM with consumed data. Knowledge transfer with refined KGs shows increased sample efficiency.

utilized to aid the design of the hysteresis comparator. We show the FOM evolution of the comparator in Figure 7.

D. DC-DC Converter

The DC-DC converter is a PWM-controlled circuit designed to step down voltage, with 44 design parameters across 22 transistors. It contains substructure types such as the error amplifier and comparator which are also found in prior circuits. To enable knowledge transfer, we incorporate KGs from the differential amplifier, hysteresis comparator, and LDO. The design targets include output power $< 15\,\mu$W and regulated voltage error < 0.02 V relative to a 0.6 V reference. As shown in Table III, `ADO-KT` consistently outperforms ADO-LLM, particularly when refined KGs are used, highlighting the benefit of multi-source knowledge transfer.

E. Key Observations

1) ***Impact of Knowledge Transfer on Design Quality and Sample Efficiency****:* Across all the circuits, incorporating KG-based knowledge transfer significantly improves design quality and sample efficiency. As shown in Tables I, II, and III, transferred knowledge enables higher FOMs and more designs that meet specifications. Figure 7 further shows faster FOM convergence, underscoring the value of knowledge reuse for sample efficient optimization.

2) ***Role of Knowledge Graph Quality****:* Optimization performance is highly sensitive to KG quality, with refined KGs yielding further improvements over their unrefined counterparts. These results affirm that (i) the effectiveness of knowledge transfer depends on the reliability of the injected knowledge, and (ii) improving knowledge representation and refinement directly enhances transfer-driven analog design.

VI. Conclusion and Future Work

Our work demonstrates the promise of structured knowledge representations for enabling knowledge transfer in analog circuit design. Moving forward, we anticipate that more advanced strategies for summarizing and distilling analog design knowledge from LLMs is critical to improving transferability. Furthermore, the development of specialized knowledge refinement mechanisms will be essential, not only to enhance consistency, but also to yield deeper qualitative insights that support interpretable and robust analog design optimization.

Acknowledgment

This material is based upon work supported by the Semiconductor Research Corporation Task No. 3160.055 through UT Dallas' Texas Analog Center of Excellence, and by the National Science Foundation under Grants No. 1956313 and No. 2334380.

References

[1] R. Zhou, P. Poechmueller, and Y. Wang, "An analog circuit design and optimization system with rule-guided genetic algorithm," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 41, no. 12, pp. 5182–5192, 2022.

[2] R. Garnett, *Bayesian Optimization*. Cambridge University Press, 2023.

[3] A. Lberni, M. A. Marktani, A. Ahaitouf, and A. Ahaitouf, "Analog circuit sizing based on evolutionary algorithms and deep learning," *Expert Systems with Applications*, vol. 237, p. 121480, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417423019826

[4] W. Lyu, F. Yang, C. Yan, D. Zhou, and X. Zeng, "Batch Bayesian optimization via multi-objective acquisition ensemble for automated analog circuit design," ser. Proceedings of Machine Learning Research, J. Dy and A. Krause, Eds., vol. 80, 2018, pp. 3306–3314.

[5] K. Settaluri, A. Haj-Ali, Q. Huang, K. Hakhamaneshi, and B. Nikolić, "Autockt: Deep reinforcement learning of analog circuit designs," *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pp. 490–495, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:210023627

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III: DCDC Converter Optimization Results

Method	Output Power ↓ (Spec: $< 15\mu$A)	Reg. Voltage Error ↓ (Spec: < 0.02V)	FOM ↑
BO	15.25	0.091	-0.073
ADO-LLM	15.21	0.041	-0.027
`ADO-KT` : Knowledge Transfer (from TSDA, Comp, LDO) without Refinement with different $\mathcal{M}_{\mathrm{KG}}$			
DeepSeek-Distill-Llama-70B	14.3	0.029	-0.014
DeepSeek-R1-Distill-Qwen-32B	16.54	0.026	-0.022
GPT-4o	15.37	0.008	0.000
Llama3-70B-instruct	11.54	0.015	0.012
Mistral-Small-24B-Instruct-2501	11.61	0.030	-0.009
Qwen2.5-32B-Instruct	16.54	0.026	-0.022
`ADO-KT` : Knowledge Transfer (from TSDA, Comp, LDO) with Refinement with different $\mathcal{M}_{\mathrm{KG}}$			
DeepSeek-Distill-Llama-70B	14.49	0.029	0.001
DeepSeek-R1-Distill-Qwen-32B	11.57	0.037	-0.007
GPT-4o	12.31	0.007	**0.015**
Llama3-70B-instruct	9.13	0.000	**0.041**
Mistral-Small-24B-Instruct-2501	12.01	0.013	**0.012**
Qwen2.5-32B-Instruct	16.54	0.026	-0.022

[6] H. Wang, K. Wang, J. Yang, L. Shen, N. Sun, H.-S. Lee, and S. Han, "Gcn-rl circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning," *2020 57th ACM/IEEE Design Automation Conference (DAC)*, pp. 1–6, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:218470023

[7] N. S. K. Somayaji, H. Hu, and P. Li, "Prioritized reinforcement learning for analog circuit optimization with design knowledge," *2021 58th ACM/IEEE Design Automation Conference (DAC)*, pp. 1231–1236, 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:243915186

[8] K. S. Ns and P. Li, "Pareto optimization of analog circuits using reinforcement learning," *ACM Transactions on Design Automation of Electronic Systems*, vol. 29, pp. 1 – 14, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:267000553

[9] Y. Yin, Y. Wang, B. Xu, and P. Li, "Ado-llm: Analog design bayesian optimization with in-context learning of large language models," 06 2024.

[10] C. Liu, Y. Liu, Y. Du, and L. Du, "Ladac: Large language model-driven auto-designer for analog circuits." [Online]. Available: https://api.semanticscholar.org/CorpusID:268259740

[11] Y. Lai, S. Lee, G. Chen, S. Poddar, M. Hu, D. Z. Pan, and P. Luo, "Analogcoder: Analog circuit design via training-free code generation," *ArXiv*, vol. abs/2405.14918, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:270045319

[12] D. V. Kochar, H. Wang, A. Chandrakasan, and X. Zhang, "Ledro: Llm-enhanced design space reduction and optimization for analog circuits," *ArXiv*, vol. abs/2411.12930, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:274149830

[13] S. Pieper, C. W. Mehling, D. Hirsch, T. Lüke, and S. Ihlenfeldt, "causalgraph: A python package for modeling, persisting and visualizing causal graphs embedded in knowledge graphs," *ArXiv*, vol. abs/2301.08490, 2023. [Online]. Available: https://api.semanticscholar.org/CorpusID:256080522

[14] I. RobertL.Logan, N. F. Liu, M. E. Peters, M. Gardner, and S. Singh, "Barack's wife hillary: Using knowledge graphs for fact-aware language modeling," *ArXiv*, vol. abs/1906.07241, 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:189998980

[15] N. Karalis, G. M. Mandilaras, and M. Koubarakis, "Extending the yago2 knowledge graph with precise geospatial knowledge," in *International Workshop on the Semantic Web*, 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:204754876

[16] S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, pp. 494–514, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:211010433

[17] A. Hogan, E. Blomqvist, M. Cochez, C. d'Amato, G. de Melo, C. Gutierrez, J. E. L. Gayo, S. Kirrane, S. Neumaier, A. Polleres, R. Navigli, A.-C. N. Ngomo, S. M. Rashid, A. Rula, L. Schmelzeisen, J. Sequeda, S. Staab, and A. Zimmermann, "Knowledge graphs," *ACM Computing Surveys (CSUR)*, vol. 54, pp. 1 – 37, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:212414739

[18] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

[19] D. Guo and et al., "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025. [Online]. Available: https://arxiv.org/abs/2501.12948

[20] OpenAI *et al.*, "Gpt-4o system card," https://arxiv.org/abs/2410.21276, 2024, accessed: 2025-04-21.

[21] M. AI *et al.*, "Meta llama 3 70b instruct model card," https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct, 2024, accessed: 2025-04-21.

[22] ——, "Mistral-small-24b-instruct-2501 model card," https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501, 2025, accessed: 2025-04-21.

[23] Q. Team *et al.*, "Qwen2.5-32b-instruct model card," https://huggingface.co/Qwen/Qwen2.5-32B-Instruct, 2024, accessed: 2025-04-21.

Authorized licensed use limited to: University of Florida. Downloaded on September 28,2026 at 03:52:41 UTC from IEEE Xplore. Restrictions apply.